SWING: Source-Reweighting-Induced Eigen-Shrinkage for Single-Run Post-hoc Domain Generalization

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Abstract

Single-run post-hoc domain generalization is attractive because it can be attached to a strong empirical-risk-minimization pipeline without retraining. Existing post-hoc methods in this regime predominantly select or average checkpoints inside one connected low-loss region, motivated by flatness, locality, or diversity. This paper studies a different remaining degree of freedom: *once a safe basin has already been found, should all directions inside that basin be trusted equally under domain shift?* We propose **SWING** (**S**ource-re**W**eighting-**I**nduced ei**N**-shrin**G**age), a single-run, post-hoc, domain-label-free method that learns a low-rank basin geometry from one dense checkpoint trajectory and then contracts only the directions that are fragile under plausible source-support reweightings. The method first constructs a connected checkpoint pool, defines a basin center and trajectory subspace, fits a local quadratic source-risk surrogate on one held-out split, adds an explicit isotropic curvature inflation, estimates a reweighting fragility operator on a disjoint split, and finally outputs a single model by minimizing the resulting surrogate inside a basin trust region; when the constraint is inactive, this reduces to generalized eigen-shrinkage of the inflated baseline. We prove a local hidden-shift upper bound in which the reweighting term appears as a square-root variance penalty, show that any inflation level $\tau \geq \sqrt{\varrho}K$ yields a valid quadratic hidden-shift certificate, derive the closed-form generalized eigen-shrinkage solution of the unconstrained inflated surrogate, give a precise comparison with scalar shrinkage inside that quadratic model, and establish a deterministic perturbation bound together with a conditional plug-in consistency statement for the unconstrained estimator. The experimental sections intentionally leave all new SWING quantitative results as **TBD**, while recording exact published benchmark context from prior work and providing the full evaluation protocol.

1 Introduction

Domain generalization (DG) asks for a predictor that performs well on an unseen target environment using only source data at training time. In practice, the DG methods that last are often the ones that preserve the base training recipe and change only how the final model is selected. This is why single-run post-hoc selection matters: one can train once, save dense checkpoints, and choose a deployable model after training has already finished.

Historically, SWAD established checkpoint averaging as a serious DG tool by improving the average DomainBed ResNet-50 accuracy of ERM from 63.3 to 66.9 across PACS, VLCS, OfficeHome, TerraIncognita, and DomainNet [4]. Later work widened that picture. EoA studies moving-average model selection within a single run and heavier ensembles of averages across runs, reporting 68.0 average with a pre-trained ResNet-50 in its strongest setting [2]. More diverse weight-averaging schemes such as DiWA also reach 68.0 in heavier multi-run, multi-model configurations [21], a pretraining-aware method combined with the same post-hoc selector reaches 68.1 [5], and the stronger

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non-post-hoc baseline ERM++ reports 68.9 on the same ResNet-50 DomainBed suite [25]. WACV 2025 further reports stronger hybrid combinations on the same suite, including DIWA+ERM++ at 69.3 and MIRO+ERM++ at 69.4 [25]. The empirical lesson is clear: the final predictor is often not the last checkpoint, and any new method now has to clear a stronger bar than the original SWAD milestone.

Yet existing post-hoc DG methods still work inside one narrow action family. They select a checkpoint, an interval, or a soup inside a connected low-loss region. That is often enough to find a safe basin, but it leaves a natural question unanswered:

> **After a safe basin has been identified, should the final model keep every direction inside that basin at full strength?**

Our answer is no. Some directions inside a good basin are strongly supported by the pooled source data; others are fragile because small source-support redistributions would swing their contribution substantially. Those fragile directions are exactly the ones that should be attenuated before deployment.

This principle leads to **SWING** (**S**ource-re**W**eighting-**I**nduced ei**N**-shrin**G**age). SWING has two stages. First, it builds a safe connected checkpoint pool, computes a basin center, and extracts a low-rank trajectory subspace. Second, inside that subspace, it fits:

- a *source-fit operator* A_0 that captures how strongly the pooled source data supports each direction,
- a *fragility operator* B that measures how much examplewise local loss slopes vary across the source support, hence how strongly nearby source reweightings can swing the direction.

It then adds an explicit isotropic inflation $A_\tau = A_0 + \tau I_r$ and minimizes the resulting surrogate inside the basin trust region. When the trust-region constraint is inactive, the edit reduces to shrinking the inflated baseline along the generalized eigenvectors of the pair (B, A_τ). The hidden-shift certificate corresponds to any $\tau \geq \sqrt{\varrho}K$; in practice, τ is an explicit inflation hyperparameter.

The central idea may be stated in one sentence:

> **Directions that swing under plausible source reweightings should not survive deployment at full strength.**

This move is still grounded in the current post-hoc DG literature. Weight averaging, mode connectivity, soups, and moving-average selectors tell us that dense trajectories often lie in a common low-loss basin [2, 9, 10, 13, 26]. Subspace-based Bayesian approximations further show that a small trajectory subspace can contain many high-performing models [14]. Influence analyses and distributionally robust optimization show that sample reweightings are a natural first-order model of nearby distribution shifts [6, 8, 16]. SWING combines these threads into a post-hoc DG method that is still simple, still single-model at test time, and more directly tied to hidden shift than isotropic averaging.

We make four contributions.

1. We introduce SWING, a single-run, post-hoc, domain-label-free DG method that performs a full-weight basin edit rather than another checkpoint selector.
2. We derive SWING from a local hidden-shift model in which target domains are source-supported χ^2-bounded reweightings of the pooled source support. This yields a square-root fragility term $\sqrt{\varrho\, z^\top B z}$ and a certificate family of quadratic surrogates indexed by an isotropic inflation parameter $\tau \geq \sqrt{\varrho}K$.
3. We prove that the unconstrained inflated SWING edit has an exact generalized eigen-shrinkage form, and that the deployed estimator inherits this form whenever the trust-region constraint is inactive. We also give a precise within-model comparison against scalar shrinkage.
4. We prove a deterministic perturbation bound and a conditional plug-in consistency statement for the unconstrained empirical estimator, and give a restricted-family target-risk certificate for the population minimizer of the chosen hidden-shift surrogate.

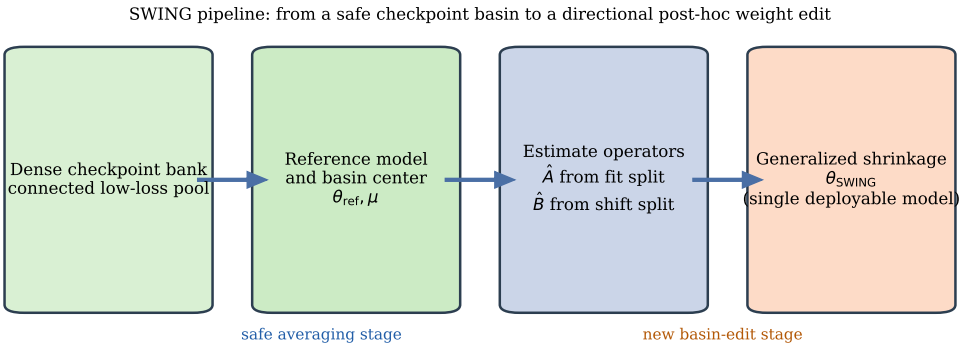


Figure 1: **SWING pipeline.** A dense checkpoint bank is reduced to a connected safe pool. That pool defines a basin center μ and a low-rank subspace U. On disjoint source splits, SWING estimates a source-fit operator $\hat{A}_0$, inflates it to $\hat{A}_\tau$, estimates a reweighting fragility operator $\hat{B}$, and outputs one deployable model by minimizing the surrogate within the basin trust region; when the constraint is inactive, this reduces to generalized eigen-shrinkage of the inflated baseline.

2 Related Work

Single-run post-hoc DG. SWAD is the canonical checkpoint-averaging reference in single-run post-hoc DG [4]. EoA studies the same basic regime from a different angle: it uses moving averages within a run for model selection and also studies heavier ensembles of averages across runs, explicitly arguing that averaging can improve the correlation between validation and test performance [2]. Published follow-ups reinforce that post-hoc averaging is genuinely strong. At the same time, the strongest published DIWA configuration is explicitly heavier, averaging models from different training runs [21], while MIRO improves the training representation and then benefits again from post-hoc selection [5]. SWING stays in the single-run deployment regime and changes the post-hoc action itself: it edits one final weight vector inside a safe basin rather than merely choosing which checkpoints to average.

Current benchmark landscape. The empirical bar has also moved. In addition to the historical SWAD/DIWA/MIRO milestones, ERM++ shows that a carefully strengthened plain baseline reaches 68.9 average on the same ResNet-50 DomainBed suite, while hybrid combinations such as DIWA+ERM++ and MIRO+ERM++ reach 69.3 and 69.4, respectively [25]. SWING is therefore not positioned against the 2021 landscape alone; it should be judged against both the post-hoc literature and stronger current baselines.

Weight averaging, connectivity, and basin geometry. Stochastic weight averaging [13], fast geometric ensembling [10], linear mode connectivity [9], and model soups [26] show that dense training trajectories often traverse connected low-loss regions. Subspace inference further shows that a small trajectory subspace can contain many strong models [14]. SWING uses these ideas as infrastructure: a connected checkpoint pool defines the basin, and a low-rank subspace provides the coordinate system for post-hoc editing.

Reweighting, influence, and local robustness. Influence functions quantify local sensitivity to sample reweightings [16]. Distributionally robust optimization studies worst-case risk over ambiguity sets such as χ^2 or other f-divergence balls [6–8]. Closely related literatures study worst-case subpopulations, fairness without observed demographics, group-robust learning through reweighting, group-robust sample reweighting via influence functions, and unlabeled feature reweighting for group robustness [12, 17–20, 23]. Anchor regression is also relevant as a classical example of a structured shift-robust penalty rather than generic regularization [22]. Tangent Prop is the classical precursor in input space: useful invariance can be enforced by controlling derivatives along nuisance directions [24]. SWING transfers this logic to the empirical distribution: its nuisance directions are nearby redistributions of the source support.

Adjacent post-training weight editing outside DG. There is recent interest in editing weights after training to improve robustness or preserve OOD performance, including WiSE-FT interpolation, robustness signatures, model-stock averaging, and spectral soup editing [1, 3, 15, 27]. These works are adjacent but different. They do not target single-run post-hoc DG with source-only held-out data, do not derive a hidden-subpopulation reweighting geometry, and do not formulate the final edit as the solution of a basin-restricted hidden-shift surrogate whose unconstrained form is generalized eigen-shrinkage.

3 Problem Setup

We work in the standard source-only DG setting. Let P denote the pooled source distribution over examples $Z = (X, Y)$. A dense training run produces checkpoints $\theta_1, \ldots, \theta_T \in \mathbb{R}^d$. The method is post-hoc: training is already over, and only the checkpoints and held-out source data are used.

Safe checkpoint pool. Let S_{pool} be a held-out source split used for pool construction. Define a source validation loss $\widehat{L}_{\text{pool}}(\theta)$ on this split. Let

$$t^\star \in \operatorname*{arg\,min}_{1 \le t \le T} \widehat{L}_{\text{pool}}(\theta_t)$$

be the best checkpoint. For checkpoints i, j, define the empirical interpolation barrier

$$\text{bar}(i, j) = \max_{\alpha \in \mathcal{G}} \widehat{L}_{\text{pool}}\left(\alpha\theta_i + (1-\alpha)\theta_j\right) - \max\left\{\widehat{L}_{\text{pool}}(\theta_i), \widehat{L}_{\text{pool}}(\theta_j)\right\},$$

where $\mathcal{G} \subset [0, 1]$ is a fixed interpolation grid. For user-chosen thresholds $\varepsilon_{\text{pool}}, \tau_{\text{pool}} > 0$, define the safe pool

$$\mathcal{C} = \left\{t : \widehat{L}_{\text{pool}}(\theta_t) \le \widehat{L}_{\text{pool}}(\theta_{t^\star}) + \varepsilon_{\text{pool}},\ \text{bar}(t, t^\star) \le \tau_{\text{pool}}\right\}.$$

This pool is intended to isolate one connected low-loss basin. The pool construction stage is not the novelty of the paper; it is the infrastructure on top of which SWING operates.

Basin coordinates. Define the basin center

$$\mu = \frac{1}{|\mathcal{C}|} \sum_{t \in \mathcal{C}} \theta_t.$$

Let $U \in \mathbb{R}^{d \times r}$ have orthonormal columns spanning the top r principal components of $\{\theta_t - \mu : t \in \mathcal{C}\}$. Every point $z \in \mathbb{R}^r$ induces a full model

$$\theta(z) = \mu + Uz.$$

The safe pool itself induces observed coordinates

$$z_t = U^\top(\theta_t - \mu), \qquad t \in \mathcal{C}.$$

For every such coordinate, we also define the projected subspace model

$$\tilde{\theta}_t = \theta(z_t) = \mu + Uz_t, \qquad t \in \mathcal{C}.$$

Disjoint source splits. Beyond S_{pool}, SWING uses two disjoint held-out source splits:

- S_{fit}, used to fit a local quadratic source-risk surrogate;
- S_{shift}, used to estimate the fragility operator.

The split separation is deliberate: it reduces direct reuse of the same source examples when fitting the source geometry and the hidden-shift geometry, although it does not by itself guarantee independence or unbiasedness once the upstream checkpoint pool and subspace have already been selected.

In theory, $\mathcal{Z}$ is any compact trust region containing the observed safe-pool coordinates. In practice, we instantiate $\mathcal{Z}$ as a box in subspace coordinates whose sides are determined by the extrema of the safe-pool projections.

4 The Method

4.1 Quadratic source-fit surrogate

Choose a local fitting neighborhood

$$\mathcal{C}_{\mathrm{loc}} = \{t \in \mathcal{C} : \|z_t\| \le \rho_{\mathrm{loc}}\}$$

for a user-chosen radius $\rho_{\mathrm{loc}} > 0$. On S_{fit}, SWING fits a local quadratic model of source risk in the basin subspace:

$$q_0(z) = c + a^\top z + \frac{1}{2} z^\top A_0 z, \qquad A_0 \in \mathbb{R}^{r\times r},\ A_0 \succ 0.$$

In practice, q_0 is estimated from the local projected-model cloud $\left\{(z_t, \tilde{\theta}_t) : t \in \mathcal{C}_{\mathrm{loc}}\right\}$ by least-squares fitting of the empirical losses

$$\widehat{L}_{\mathrm{fit}}(\tilde{\theta}_t) \approx c + a^\top z_t + \frac{1}{2} z_t^\top A_0 z_t.$$

The unique minimizer of the quadratic fit-only surrogate is

$$z_{\mathrm{fit}} = -A_0^{-1} a.$$

This is the source-fit solution in the basin coordinate system.

4.2 Fragility operator from source reweightings

Fix an example $Z_i \in S_{\mathrm{shift}}$. Let $\phi_i \in \mathbb{R}^r$ denote the local first-order loss slope of that example in the basin subspace. Formally, in the theory we assume

$$\ell\left(\theta(z); Z_i\right) = \ell\left(\theta(0); Z_i\right) + \phi_i^\top z + r_i(z),$$

with a controlled quadratic remainder $r_i(z)$.

The fragility operator is the covariance of these examplewise slopes:

$$B = \mathrm{Cov}(\phi) = \mathbb{E}\left[(\phi - \mathbb{E}[\phi])(\phi - \mathbb{E}[\phi])^\top\right].$$

Intuitively:

- if a direction has small variance in examplewise slopes, then nearby source reweightings do not change its preferred update much;
- if a direction has large variance, then small hidden redistributions of the source support can swing its utility sharply.

In practice, ϕ_i is estimated locally around $z = 0$, not from the entire checkpoint cloud. For each example Z_i, SWING fits a local quadratic model on $\mathcal{C}_{\mathrm{loc}}$,

$$\ell(\tilde{\theta}_t; Z_i) \approx \alpha_i + \hat{\phi}_i^\top z_t + \frac{1}{2} z_t^\top \hat{H}_i z_t, \qquad t \in \mathcal{C}_{\mathrm{loc}},$$

and takes the fitted linear coefficient $\hat{\phi}_i$ as the estimated derivative at the basin center. The empirical fragility operator $\hat{B}$ is then the sample covariance of these local slope estimates.

4.3 The SWING objective

For any inflation level $\tau \ge 0$, define

$$A_\tau = A_0 + \tau I_r.$$

The derivation in Section 5 yields a *family* of quadratic hidden-shift surrogates

$$J_{\lambda,\tau}(z) = a^\top z + \frac{1}{2} z^\top A_\tau z + \frac{\lambda}{2} z^\top B z, \qquad \lambda > 0,\ \tau \ge 0.$$

The first two terms combine source fit with isotropic curvature inflation. The third term penalizes directions whose first-order contribution is unstable under plausible source-support reweightings.

In the hidden-shift theory, every $\tau \geq \sqrt{\varrho}K$ is certificate-compatible; in practice τ is an explicit hyperparameter.

The analyzed population surrogate minimizer is

$$z^{\mathcal{Z}}_{\lambda,\tau} \in \arg\min_{z\in\mathcal{Z}} J_{\lambda,\tau}(z).$$

Whenever the unconstrained minimizer lies inside $\mathcal{Z}$, the constrained and unconstrained solutions coincide.

Given empirical plug-in estimates $(\hat{a}, \hat{A}_0, \hat{B})$, define

$$\hat{A}_\tau = \hat{A}_0 + \tau I_r, \qquad \hat{J}_{\lambda,\tau}(z) = \hat{a}^\top z + \frac{1}{2} z^\top \hat{A}_\tau z + \frac{\lambda}{2} z^\top \hat{B} z.$$

The practical returned estimator is the constrained empirical plug-in minimizer

$$\hat{z}^{\mathcal{Z}}_{\lambda,\tau} \in \arg\min_{z\in\mathcal{Z}} \hat{J}_{\lambda,\tau}(z).$$

Let

$$z_\tau = -A_\tau^{-1} a.$$

Then

$$J_{\lambda,\tau}(z) = \text{const} + \frac{1}{2}(z - z_\tau)^\top A_\tau (z - z_\tau) + \frac{\lambda}{2} z^\top B z.$$

Therefore SWING should be interpreted as a *basin edit*: it does not search for a new basin, but contracts an inflated local baseline along fragile directions inside a basin that was already judged safe by the first-stage pool construction.

4.4 Generalized eigen-shrinkage

Because $A_\tau \succ 0$ and $B \succeq 0$, there exists an A_τ-orthonormal generalized eigenbasis $V = [v_1, \ldots, v_r]$ and nonnegative generalized eigenvalues $\nu_1, \ldots, \nu_r$ such that

$$B v_j = \nu_j A_\tau v_j, \qquad V^\top A_\tau V = I.$$

Writing

$$z_\tau = \sum_{j=1}^{r} c_j v_j, \qquad c_j = v_j^\top A_\tau z_\tau,$$

the unconstrained SWING edit is

$$z^{\text{unc}}_{\lambda,\tau} = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j.$$

Thus every generalized coordinate is shrunk by a factor

$$\beta_j(\lambda) = \frac{1}{1 + \lambda \nu_j}.$$

Directions with larger fragility-to-fit ratio ν_j receive stronger attenuation.

5 Theory

This section derives the SWING objective from a local hidden-shift model and proves its main properties. The assumptions are deliberately explicit. The theory is local and subspace-restricted; it does not claim that arbitrary deep networks are globally quadratic in weight space.

Algorithm 1 SWING

Require: Dense checkpoint bank $\{\theta_t\}_{t=1}^{T}$, held-out source splits $S_{\text{pool}}, S_{\text{fit}}, S_{\text{shift}}$, rank r, thresholds $\varepsilon_{\text{pool}}, \tau_{\text{pool}}$, local radius ρ_{loc}, regularization λ, inflation τ.

Select the safe checkpoint pool $\mathcal{C}$ using source-fit thresholding and interpolation barriers on S_{pool}.
Compute the basin center $\mu = |\mathcal{C}|^{-1} \sum_{t \in \mathcal{C}} \theta_t$.
Compute $U \in \mathbb{R}^{d \times r}$ from the top r principal directions of $\{\theta_t - \mu : t \in \mathcal{C}\}$.
Form checkpoint coordinates $z_t = U^\top(\theta_t - \mu)$, $t \in \mathcal{C}$.
Define projected subspace models $\tilde{\theta}_t = \mu + U z_t$, $t \in \mathcal{C}$.
Keep the local neighborhood $\mathcal{C}_{\text{loc}} = \{t \in \mathcal{C} : \|z_t\| \le \rho_{\text{loc}}\}$.
Fit the local quadratic source-risk surrogate $q_0(z) = c + a^\top z + \frac{1}{2} z^\top A_0 z$ on S_{fit} using the projected models $\tilde{\theta}_t$, $t \in \mathcal{C}_{\text{loc}}$.
Form the inflated fit operator $\hat{A}_\tau = \hat{A}_0 + \tau I_r$.
Fit local examplewise quadratic models on S_{shift} over the projected models $\tilde{\theta}_t$, $t \in \mathcal{C}_{\text{loc}}$, extract linear coefficients $\hat{\phi}_i$, and set $\hat{B} = \widehat{\text{Cov}}(\hat{\phi})$.
Define the empirical plug-in objective $\hat{J}_{\lambda,\tau}(z) = \hat{a}^\top z + \frac{1}{2} z^\top \hat{A}_\tau z + \frac{\lambda}{2} z^\top \hat{B} z$.
Solve the generalized eigenproblem $\hat{B} v_j = \nu_j \hat{A}_\tau v_j$, $V^\top \hat{A}_\tau V = I$.
Compute $\hat{z}_\tau = -\hat{A}_\tau^{-1} \hat{a}$ and coefficients $\hat{c}_j = v_j^\top \hat{A}_\tau \hat{z}_\tau$.
Set the unconstrained empirical edit $\hat{z}^{\text{unc}}_{\lambda,\tau} = \sum_{j=1}^{r} \frac{\hat{c}_j}{1+\lambda\nu_j} v_j$.
If $\hat{z}^{\text{unc}}_{\lambda,\tau} \in \mathcal{Z}$, set $\hat{z}^{\mathcal{Z}}_{\lambda,\tau} = \hat{z}^{\text{unc}}_{\lambda,\tau}$; otherwise solve $\hat{z}^{\mathcal{Z}}_{\lambda,\tau} \in \arg\min_{z \in \mathcal{Z}} \hat{J}_{\lambda,\tau}(z)$ inside the chosen trust region.
return $\theta_{\text{SWING}} = \mu + U \hat{z}^{\mathcal{Z}}_{\lambda,\tau}$.

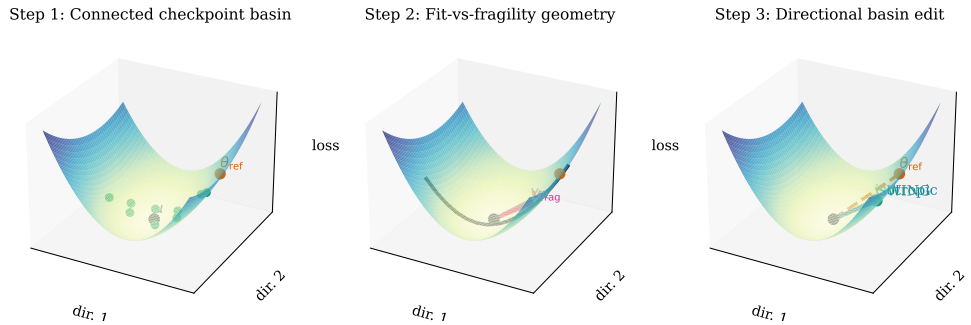


Figure 2: **Three-dimensional intuition for SWING.** Left: dense checkpoints define a connected low-loss basin and a center μ. Middle: the basin subspace supplies the coordinates in which the local fit, isotropic inflation, and fragility are estimated. Right: SWING contracts the inflated baseline anisotropically, suppressing high-fragility directions more than low-fragility directions.

5.1 Assumptions

Assumption 1 (Local quadratic source-fit model)**.** *There exists a compact set $\mathcal{Z} \subset \mathbb{R}^r$ containing 0 and the observed checkpoint coordinates, and a positive definite matrix $A_0 \in \mathbb{R}^{r \times r}$ and vector $a \in \mathbb{R}^r$ such that the pooled source excess risk on the basin subspace satisfies*

$$L_P(z) - L_P(0) = a^\top z + \frac{1}{2} z^\top A_0 z \qquad \textit{for all } z \in \mathcal{Z}.$$

Moreover $L_P(z)$ and $L_Q(z)$ are finite for all $z \in \mathcal{Z}$ and all $Q \in \mathcal{Q}(\varrho)$.

Assumption 2 (Examplewise local linearization with quadratic remainder)**.** *There exist measurable $\phi : \mathsf{Z} \to \mathbb{R}^r$ and $\kappa : \mathsf{Z} \to [0, \infty)$ such that for every example $Z \in \mathsf{Z}$ and every $z \in \mathcal{Z}$,*

$$\ell(\theta(z); Z) = \ell(\theta(0); Z) + \phi(Z)^\top z + r(Z, z),$$

where the remainder satisfies

$$|r(Z, z)| \le \frac{\kappa(Z)}{2} \|z\|^2 \qquad \textit{for all } z \in \mathcal{Z}.$$

Moreover $\mathbb{E}_P[\|\phi(Z)\|^2] < \infty$ *and* $\mathbb{E}_P[\kappa(Z)^2] \le K^2 < \infty$.

Assumption 3 (Source-supported local target shift)**.** *The target distribution Q is absolutely continuous with respect to P and belongs to the local χ^2-style ambiguity family*

$$\mathcal{Q}(\varrho) = \left\{ Q : \frac{dQ}{dP} = 1 + h,\ 1 + h \ge 0 \ P\text{-a.s.},\ \mathbb{E}_P[h] = 0,\ \mathbb{E}_P[h^2] \le \varrho \right\}$$

for some radius $\varrho \ge 0$.

Assumption 3 encodes the source-supported hidden-subpopulation view: unseen domains are not arbitrary, but are local redistributions of source support mass. Assumption 2 makes the reweighting effect observable through the examplewise slope covariance. Assumption 1 isolates the source-fit geometry inside a single connected basin.

5.2 Deriving the fragility term

Theorem 1 (Local hidden-shift upper bound)**.** *Under Assumptions 1–3, define*

$$\bar{\phi} = \mathbb{E}_P[\phi(Z)], \qquad B = \mathrm{Cov}_P(\phi(Z)) = \mathbb{E}_P\left[(\phi(Z) - \bar{\phi})(\phi(Z) - \bar{\phi})^\top\right].$$

Then for every $z \in \mathcal{Z}$ and every $Q \in \mathcal{Q}(\varrho)$,

$$L_Q(z) \le L_Q(0) + a^\top z + \frac{1}{2} z^\top A_0 z + \sqrt{\varrho\, z^\top B z} + \frac{\sqrt{\varrho}\, K}{2} \|z\|^2.$$

The theorem shows where the fragility term comes from. The quantity $z^\top B z$ is the variance of the first-order examplewise slope in direction z. If that variance is large, local source reweightings can swing the contribution of z strongly; if it is small, the direction is reweighting-stable.

Corollary 1 (Quadratic SWING surrogate)**.** *Under the assumptions of Theorem 1, for every $\lambda > 0$ and every $\tau \ge \sqrt{\varrho} K$,*

$$L_Q(z) \le L_Q(0) + \frac{\varrho}{2\lambda} + a^\top z + \frac{1}{2} z^\top A_\tau z + \frac{\lambda}{2} z^\top B z \qquad \textit{for all } z \in \mathcal{Z},$$

where

$$A_\tau = A_0 + \tau I_r.$$

Corollary 1 is the exact bridge from the hidden-shift model to SWING: for each fixed $\lambda > 0$, Young's inequality yields a quadratic upper bound, and λ controls the tradeoff between the additive constant and the fragility penalty. The hidden-shift certificate remains valid for any isotropic inflation $\tau \ge \sqrt{\varrho} K$.

5.3 Closed-form shrinkage solution

Theorem 2 (Generalized eigen-shrinkage)**.** *Let*

$$J_{\lambda,\tau}(z) = a^\top z + \frac{1}{2} z^\top A_\tau z + \frac{\lambda}{2} z^\top B z, \qquad A_\tau = A_0 + \tau I_r,\ A_\tau \succ 0,\ B \succeq 0,\ \lambda > 0,\ \tau \ge 0.$$

Define the inflated baseline

$$z_\tau = -A_\tau^{-1} a.$$

Then:

1. *The unconstrained problem $\min_{z \in \mathbb{R}^r} J_{\lambda,\tau}(z)$ is strongly convex and has the unique minimizer*

$$z_{\lambda,\tau}^{\mathrm{unc}} = -(A_\tau + \lambda B)^{-1} a = (A_\tau + \lambda B)^{-1} A_\tau z_\tau.$$

2. *If $z_{\lambda,\tau}^{\mathrm{unc}} \in \mathcal{Z}$, then $z_{\lambda,\tau}^{\mathcal{Z}} = z_{\lambda,\tau}^{\mathrm{unc}}$. In that case, if $V = [v_1, \dots, v_r]$ is an A_τ-orthonormal generalized eigenbasis satisfying*

$$B v_j = \nu_j A_\tau v_j, \qquad V^\top A_\tau V = I, \qquad \nu_j \ge 0,$$

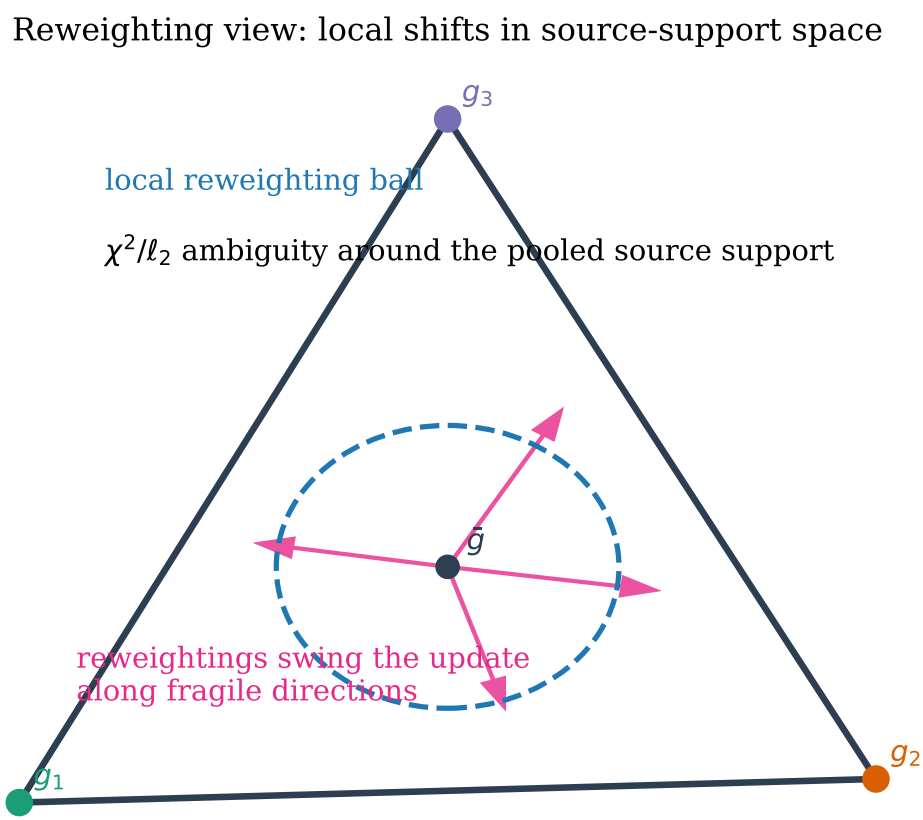


Figure 3: **Why reweighting induces a fragility operator.** Nearby target domains are modeled as local redistributions of source-support mass. Examplewise slope vectors that fan out strongly under these redistributions produce a large covariance B, so directions aligned with them are shrunk more aggressively by SWING.

and

$$z_\tau = \sum_{j=1}^{r} c_j v_j, \qquad c_j = v_j^\top A_\tau z_\tau,$$

then

$$z_{\lambda,\tau}^{\mathcal{Z}} = z_{\lambda,\tau}^{\mathrm{unc}} = \sum_{j=1}^{r} \frac{c_j}{1+\lambda\nu_j} v_j.$$

Theorem 2 is the central algebraic result of the paper. It shows that SWING is not merely a heuristic penalty. Whenever the unconstrained minimizer stays inside the chosen basin trust region, the analyzed population solution is an exact contraction of the inflated baseline, with one shrinkage factor per generalized fragility level.

5.4 Why anisotropy matters

The next comparison result formalizes an important point. If generalized fragility levels are heterogeneous, then one global scalar shrinkage factor is not enough inside the quadratic model.

Proposition 1 (Comparison to scalar shrinkage)**.** *Under the assumptions of Theorem 2, consider the scalar shrinkage family*

$$\mathcal{F}_{\mathrm{sc}} = \{\alpha z_\tau : \alpha \in \mathbb{R}\}.$$

Then

$$\min_{\alpha\in\mathbb{R}} J_{\lambda,\tau}(\alpha z_\tau) \geq J_{\lambda,\tau}(z_{\lambda,\tau}^{\mathrm{unc}}).$$

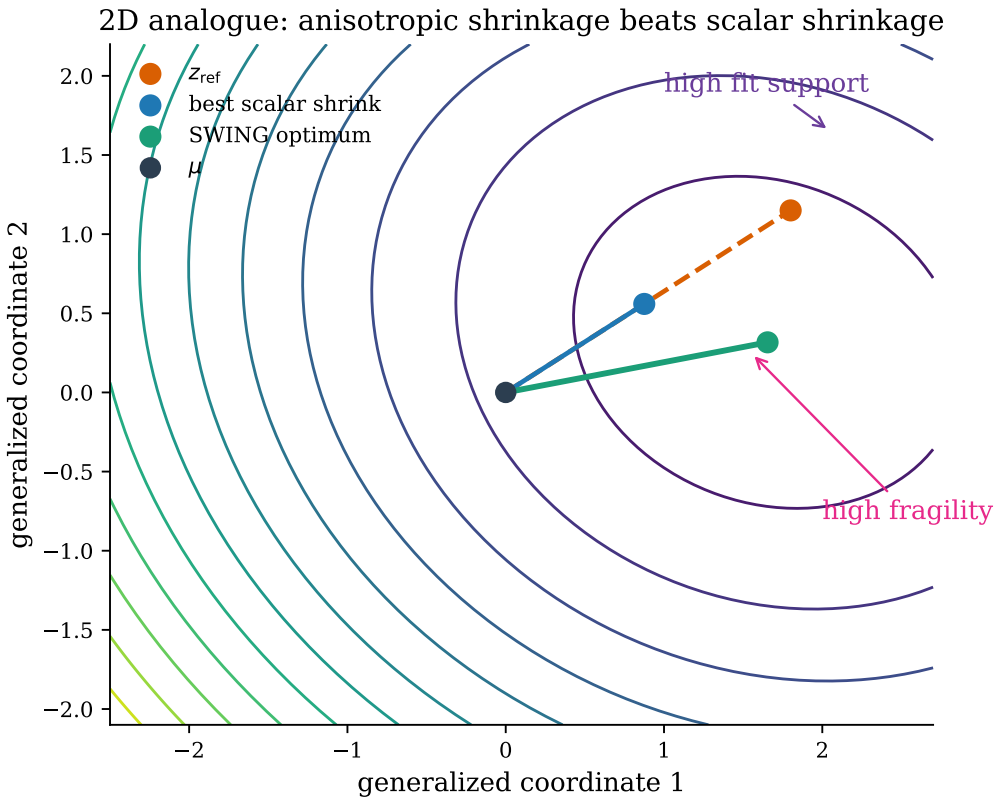


Figure 4: **Two-dimensional analogue of Proposition 1.** The inflated baseline z_τ lies away from the basin center μ. The best scalar shrinkage moves only along the dashed ray. The SWING optimum follows the anisotropic geometry defined by A_τ and B, producing a lower robust surrogate value whenever fragility is direction-dependent.

Moreover equality holds if and only if the generalized eigenvalues ν_j are constant on the support of z_τ, that is,

$$\nu_j = \nu_k \qquad \text{for all } j, k \text{ such that } c_j \neq 0 \text{ and } c_k \neq 0.$$

Consequently, whenever z_τ has nonzero mass on at least two distinct generalized fragility levels, anisotropic shrinkage is strictly better than the best scalar shrinkage.

Proposition 1 records the simplest exact comparison that motivates anisotropy in the quadratic surrogate. Scalar shrinkage is optimal only in the degenerate case where all active directions have the same fragility-to-fit ratio.

5.5 Certificate improvement and conditional stability

Proposition 2 (Exact certificate gain over the inflated baseline)**.** *Under the assumptions of Theorem 2,*

$$J_{\lambda,\tau}(z_\tau) - J_{\lambda,\tau}(z^{\text{unc}}_{\lambda,\tau}) = \frac{1}{2}(z_\tau - z^{\text{unc}}_{\lambda,\tau})^\top (A_\tau + \lambda B)(z_\tau - z^{\text{unc}}_{\lambda,\tau}).$$

In particular,

$$J_{\lambda,\tau}(z^{\text{unc}}_{\lambda,\tau}) \leq J_{\lambda,\tau}(z_\tau),$$

with equality if and only if $Bz_\tau = 0$.

Proposition 2 quantifies the exact improvement that SWING certifies over the inflated baseline. If that baseline already lies in the kernel of the fragility operator, there is nothing to shrink; otherwise the robust surrogate is improved strictly.

Assumption 4 (Conditional local identification for plug-in estimation)**.** *For the consistency discussion only, condition on a deterministic asymptotic sequence of upstream-selected basin families*

$(\mu_n, U_n, \mathcal{C}_n, \mathcal{C}_{\mathrm{loc},n}, \mathcal{Z}_n)$. *Assume that* 0 *is an interior point of every local coordinate design, that the local polynomial Gram matrix formed by the intercept, linear monomials, and quadratic monomials over* $\mathcal{C}_{\mathrm{loc},n}$ *is uniformly well conditioned, and that the fill distance of the local design around* 0 *tends to zero along the sequence. Assume further that the source-risk function and the examplewise losses restricted to each basin family are three times continuously differentiable with uniformly bounded third derivatives on a neighborhood of* 0.

Assumption 4 is not used in the algebraic population certificate. It is only the standard local-polynomial regularity needed to interpret $(\hat{a}, \hat{A}_0, \hat{\phi}_i)$ as estimators of derivatives at $z = 0$. Turning that identification statement into $\hat{A}_0 \to A_0$ and $\hat{B} \to B$ also requires the usual tuning and triangular-array conditions for local polynomial regression. A representative regime is

$$|S_{\mathrm{fit},n}|,\ |S_{\mathrm{shift},n}|,\ |\mathcal{C}_{\mathrm{loc},n}| \to \infty, \qquad \eta_{A,n},\ \eta_{H,n} \to 0,$$

while the local design matrices remain uniformly well conditioned along the sequence. The consistency discussion below is therefore explicitly *conditional* on that deterministic basin-family sequence chosen upstream.

Theorem 3 (Deterministic perturbation bound)**.** *Let*

$$M = A_\tau + \lambda B, \qquad \hat{A}_\tau = \hat{A}_0 + \tau I_r, \qquad \hat{M} = \hat{A}_\tau + \lambda \hat{B}, \qquad m = \lambda_{\min}(M).$$

Suppose $m > 0$ *and define*

$$\delta = \left\| \hat{A}_\tau - A_\tau \right\| + \lambda \left\| \hat{B} - B \right\|.$$

If $\delta < m$*, then* $\hat{M}$ *is positive definite and the empirical solution*

$$\hat{z}_{\lambda,\tau} = -\hat{M}^{-1}\hat{a}$$

is well defined. Moreover

$$\left\| \hat{z}_{\lambda,\tau} - z^{\mathrm{unc}}_{\lambda,\tau} \right\| \le \frac{\|\hat{a} - a\| + \delta \left\| z^{\mathrm{unc}}_{\lambda,\tau} \right\|}{m - \delta}.$$

Consequently, if $\|\hat{a} - a\| \to 0$*,* $\hat{A}_0 \to A_0$ *in operator norm, and* $\hat{B} \to B$ *in operator norm, then* $\hat{z}_{\lambda,\tau} \to z^{\mathrm{unc}}_{\lambda,\tau}$.

Theorem 3 is the paper's main convergence statement. It is a deterministic continuity bound: conditional on consistent local estimators for a, A_0, and B under Assumption 4, the resulting unconstrained SWING estimator is also consistent. It is not a consistency theorem for the full random basin-selection pipeline, and it covers the deployed edit only on the event that the trust-region constraint is inactive.

Corollary 2 (Restricted-family target certificate)**.** *Under Assumptions 1–3, fix any* $\tau \ge \sqrt{\varrho}K$ *and let*

$$z^{\mathcal{Z}}_{\lambda,\tau} \in \operatorname*{arg\,min}_{z \in \mathcal{Z}} J_{\lambda,\tau}(z).$$

Then for every $Q \in \mathcal{Q}(\varrho)$ *and every* $z \in \mathcal{Z}$*,*

$$L_Q(z^{\mathcal{Z}}_{\lambda,\tau}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_{\lambda,\tau}(z^{\mathcal{Z}}_{\lambda,\tau}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_{\lambda,\tau}(z).$$

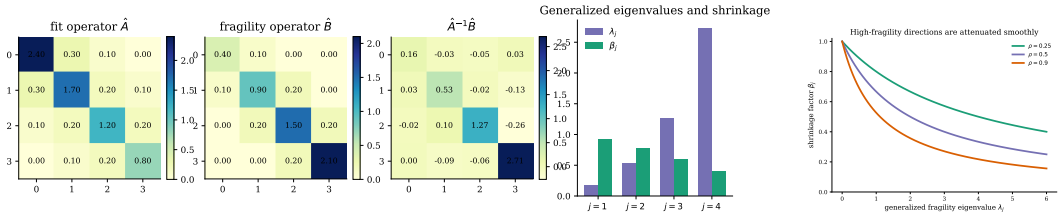
In particular, the population minimizer of the chosen hidden-shift surrogate attains the smallest value of that quadratic certificate inside the chosen basin family.

6 Practical Estimation Details

Estimating the quadratic fit surrogate. Given local coordinates z_t, projected models $\tilde{\theta}_t = \mu + U z_t$, $t \in \mathcal{C}_{\mathrm{loc}}$, and fit-split losses $\widehat{L}_{\mathrm{fit}}(\tilde{\theta}_t)$, we estimate $(\hat{c}, \hat{a}, \hat{A}_0)$ by solving

$$\min_{c,a,A_0 = A_0^\top} \sum_{t \in \mathcal{C}_{\mathrm{loc}}} \left(\widehat{L}_{\mathrm{fit}}(\tilde{\theta}_t) - c - a^\top z_t - \frac{1}{2} z_t^\top A_0 z_t \right)^2 + \eta_A \|A_0\|_F^2$$

subject to $\hat{A}_0 \succeq m_A I_r$ for a small ridge floor $m_A > 0$. We then set $\hat{A}_\tau = \hat{A}_0 + \tau I_r$. This makes the local source-fit surrogate strictly convex and numerically stable.



(a) Inflated fit and fragility operators, their generalized geometry, and the resulting spectrum.

(b) Shrinkage path as fragility rises.

Figure 5: **Operator-level intuition for SWING.** The inflated fit operator $\hat{A}_\tau$ says which directions the source data trusts after isotropic certification inflation. The fragility operator $\hat{B}$ says which directions swing under source-support redistributions. Their generalized eigensystem defines the contraction geometry.

Estimating the slope covariance. For each example $Z_i \in S_{\text{shift}}$, let $\ell_{it} = \ell(\tilde{\theta}_t; Z_i)$. We estimate the examplewise slope vector $\hat{\phi}_i$ from the local quadratic model

$$\ell_{it} \approx \alpha_i + \hat{\phi}_i^\top z_t + \frac{1}{2} z_t^\top \hat{H}_i z_t, \qquad t \in \mathcal{C}_{\text{loc}},$$

using ridge regression over $t \in \mathcal{C}_{\text{loc}}$. The fitted linear coefficient is the estimated derivative at $z = 0$. Then

$$\hat{B} = \frac{1}{|S_{\text{shift}}|} \sum_{i \in S_{\text{shift}}} (\hat{\phi}_i - \bar{\phi})(\hat{\phi}_i - \bar{\phi})^\top, \qquad \bar{\phi} = \frac{1}{|S_{\text{shift}}|} \sum_i \hat{\phi}_i.$$

Local design and identifiability. The quadratic fit is informative only if the local checkpoint design contains enough variation. Concretely, the design matrix formed by the intercept, the r linear monomials, and the $\frac{r(r+1)}{2}$ unique quadratic monomials over $\mathcal{C}_{\text{loc}}$ must be well conditioned; otherwise the ridge floor dominates the estimate and the recovered curvature is largely arbitrary. The same requirement applies to the per-example local quadratic regressions used to estimate $\hat{\phi}_i$. These are standard local-polynomial identifiability conditions rather than consequences of split separation.

Computational complexity. If d is the ambient parameter dimension, $r \ll d$ is the subspace rank, $M = |\mathcal{C}|$ is the safe-pool size, and $n_s = |S_{\text{shift}}|$, then the dominant costs are:

- subspace extraction from M checkpoints;
- one local quadratic fit in r variables for the source surrogate;
- n_s local quadratic regressions in r variables to estimate slopes;
- one $r \times r$ generalized eigendecomposition.

The final deployment cost remains unchanged from ordinary single-model inference because SWING outputs one weight vector.

7 Experimental Protocol

All new empirical results for SWING are intentionally left as **TBD**. This section records the protocol, published benchmark context, and the exact ablations the eventual empirical study should run.

7.1 Benchmarks and training regime

We will evaluate on the standard DomainBed suite [11]:

- PACS,
- VLCS,
- OfficeHome,

Method	PACS	VLCS	OfficeHome	TerraInc.	DomainNet	Avg.
ERM [4]	85.5	77.5	66.5	46.1	40.9	63.3
SWAD [4]	88.1	79.1	70.6	50.0	46.5	66.9
ERM++ [25]	89.8	78.0	74.7	51.2	50.8	68.9

Table 1: Standalone and direct single-model benchmark context from prior work. ERM++ is not a post-hoc method, but it is a strong current baseline and materially raises the empirical bar.

Setting	Method	Avg.
Strongest EoA setting	EoA [2]	68.0
Heavier multi-run post-hoc	DIWA [21]	68.0
Hybrid post-hoc	MIRO + SWAD [5]	68.1
Hybrid on stronger baseline	DIWA + ERM++ [25]	69.3
Hybrid on stronger baseline	MIRO + ERM++ [25]	69.4

Table 2: Heavier or hybrid benchmark context from prior work. These rows are useful reference points, but they are not the pure single-run post-hoc regime targeted by SWING.

- TerraIncognita,
- DomainNet.

The intended baseline regime is the standard ResNet-50 single-run post-hoc setting anchored by published SWAD comparisons. EoA is included only as adjacent published context because it reports both within-run moving-average selection and heavier across-run ensemble variants. Heavier multi-run methods such as DIWA and ensemble variants of EoA are retained only as broader published context, not as the pure single-run reference regime. The checkpoint bank should be dense enough to resolve one late connected basin. All post-hoc selection and shrinkage must use only source-held-out data and must not use source-domain identities.

7.2 Published benchmark context

Table 1 records exact published numbers for the main standalone baselines and direct single-model references. Table 2 then summarizes heavier averaging settings and stronger hybrid combinations. These are not new SWING results; they define the empirical landscape the method must eventually enter.

7.3 Planned main table

7.4 Required ablations

The empirical study should include at least the following ablations.

1. **No fragility term:** set $B = 0$ and recover the inflated baseline z_τ.
2. **Scalar shrinkage:** replace generalized shrinkage with the best one-parameter interpolation toward μ.
3. **No safe pool:** build U from a broader late trajectory without the barrier screen.
4. **Rank sensitivity:** vary r.
5. **Regularization sensitivity:** vary λ.
6. **Inflation sensitivity:** vary τ.
7. **Split overlap test:** compare disjoint versus reused fit/shift splits.

8 Discussion

What SWING is and is not. SWING is not another global soup optimizer. It is also not a last-layer repair method, not a multi-run averaging scheme, and not a target-time adaptation method. It is a full-weight basin edit that acts only after a safe post-hoc basin has already been found.

Method	PACS	VLCS	OfficeHome	TerraInc.	DomainNet	Avg.
ERM	85.5	77.5	66.5	46.1	40.9	63.3
SWAD	88.1	79.1	70.6	50.0	46.5	66.9
ERM++	89.8	78.0	74.7	51.2	50.8	68.9
SWING	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table 3: Planned primary comparison in the pure single-run post-hoc regime, together with the strong standalone baseline ERM++. Heavier and hybrid contexts are summarized separately in Table 2. All SWING entries are intentionally left as **TBD** until experiments are run.

Variant	Avg. Acc.
SWING	**TBD**
No fragility term ($B = 0$)	**TBD**
Best scalar shrinkage	**TBD**
No safe pool screen	**TBD**
Small rank r	**TBD**
Large rank r	**TBD**
Small λ	**TBD**
Large λ	**TBD**
Small τ	**TBD**
Large τ	**TBD**

Table 4: Planned ablation table. All new numbers remain **TBD**.

Why the theory is local. The paper's theory is intentionally restricted to a connected low-rank basin family. That is a strength, not a weakness. The whole operational point of post-hoc single-run DG is that one dense trajectory often already contains a safe region; the scientifically honest question is what can be proved *inside that region*.

Limitations. The theory assumes a source-supported local shift model. If the target domain introduces genuinely new support, then source-reweighting fragility cannot fully characterize deployment risk. The quadratic source-fit model is also local, and the regression estimators for A_0 and B can become noisy if the safe pool is too small or too narrow or if the local design is poorly conditioned. The certificate interpretation additionally depends on choosing $\tau \geq \sqrt{\varrho}K$, which is not directly observable in practice. Finally, the eventual empirical study must verify that the gains are not captured by simpler scalar shrinkage or by the isotropic inflation alone.

9 Conclusion

SWING is a simple claim made operational: once a good checkpoint basin has been found, not all directions inside that basin should be trusted equally. Directions whose contribution swings under plausible source-support reweightings should be shrunk. The resulting method stays single-run, post-hoc, domain-label-free, and single-model at test time. Its math is explicit: a hidden-shift upper bound yields a square-root fragility term, a family of isotropically inflated quadratic certificates, the exact generalized eigen-shrinkage form of the unconstrained inflated edit, a clean comparison against scalar shrinkage inside that quadratic model, and a deterministic perturbation theory that upgrades to consistency for the unconstrained estimator under consistent local operator estimation. The empirical study remains to be run; the paper, figures, proofs, and protocol are ready.

References

[1] Alireza Abdollahpoorrostam, Nikolaos Dimitriadis, Adam Hazimeh, and Pascal Frossard. Model soups need only one ingredient. *arXiv preprint arXiv:2602.09689*, 2026.

[2] Devansh Arpit, Huan Wang, Yingbo Zhou, and Caiming Xiong. Ensemble of averages: Improving model selection and boosting performance in domain generalization. In *Advances in Neural Information Processing Systems*, 2022.

[3] Ruisi Cai, Zhenyu Zhang, and Zhangyang Wang. Robust weight signatures: Gaining robustness as easy as patching weights? In *International Conference on Machine Learning*, 2023.

[4] Junbum Cha, Sanghyuk Chun, Kyungjae Lee, Han-Cheol Cho, Seunghyun Park, Yunsung Lee, and Sungrae Park. SWAD: Domain generalization by seeking flat minima. In *Advances in Neural Information Processing Systems*, 2021.

[5] Junbum Cha, Kyungjae Lee, Sungrae Park, and Sanghyuk Chun. Domain generalization by mutual-information regularization with pre-trained models. In *European Conference on Computer Vision*, 2022.

[6] John Duchi and Hongseok Namkoong. Variance-based regularization with convex objectives. *Journal of Machine Learning Research*, 20(68):1–55, 2019.

[7] John C. Duchi, Peter W. Glynn, and Hongseok Namkoong. Stochastic gradient methods for distributionally robust optimization with f-divergences. In *Advances in Neural Information Processing Systems*, 2016.

[8] John C. Duchi, Peter W. Glynn, and Hongseok Namkoong. Learning models with uniform performance via distributionally robust optimization. *The Annals of Statistics*, 49(3):1378–1406, 2021.

[9] Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020.

[10] Timur Garipov, Pavel Izmailov, Dmitrii Podoprikhin, Dmitry P. Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In *Advances in Neural Information Processing Systems*, 2018.

[11] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2021.

[12] Tatsunori B. Hashimoto, Megha Srivastava, Hongseok Namkoong, and Percy Liang. Fairness without demographics in repeated loss minimization. In *International Conference on Machine Learning*, 2018.

[13] Pavel Izmailov, Dmitrii Podoprikhin, Timur Garipov, Dmitry P. Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Conference on Uncertainty in Artificial Intelligence*, 2018.

[14] Pavel Izmailov, Wesley J. Maddox, Polina Kirichenko, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Subspace inference for bayesian deep learning. In *Proceedings of The 35th Uncertainty in Artificial Intelligence Conference*, volume 115, pages 1169–1179, 2020.

[15] Dong-Hwan Jang, Sangdoo Yun, and Dongyoon Han. Model stock: All we need is just a few fine-tuned models. In *European Conference on Computer Vision*, 2024.

[16] Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, 2017.

[17] Preethi Lahoti, Krishna P. Gummadi, and Gerhard Weikum. Fairness without demographics through adversarially reweighted learning. In *Advances in Neural Information Processing Systems*, 2020.

[18] Mike Li, Hongseok Namkoong, and Shangzhou Xia. Evaluating model performance under worst-case subpopulations. In *Advances in Neural Information Processing Systems*, 2021.

[19] Rui Qiao, Zhaoxuan Wu, Jingtan Wang, Pang Wei Koh, and Bryan Kian Hsiang Low. Group-robust sample reweighting for subpopulation shifts via influence functions. In *International Conference on Learning Representations*, 2025.

[20] Shikai Qiu, Andres Potapczynski, Pavel Izmailov, and Andrew Gordon Wilson. Simple and fast group robustness by automatic feature reweighting. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 28448–28467. PMLR, 2023. URL `https://proceedings.mlr.press/v202/qiu23c.html`.

[21] Alexandre Ramé, Matthieu Kirchmeyer, Thibaud Rahier, Alain Rakotomamonjy, Patrick Gallinari, and Matthieu Cord. Diverse weight averaging for out-of-distribution generalization. In *Advances in Neural Information Processing Systems*, 2022.

[22] Dominik Rothenhäusler, Nicolai Meinshausen, Peter Bühlmann, and Jonas Peters. Anchor regression: Heterogeneous data meet causality. *Journal of the Royal Statistical Society: Series B*, 83(2):215–246, 2021.

[23] Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations*, 2020.

[24] Patrice Y. Simard, Bernard Victorri, Yann LeCun, and John S. Denker. Tangent prop: A formalism for specifying selected invariances in an adaptive network. In *Advances in Neural Information Processing Systems*, 1992.

[25] Piotr Teterwak, Kuniaki Saito, Theodoros Tsiligkaridis, Kate Saenko, and Bryan A. Plummer. ERM++: An improved baseline for domain generalization. In *IEEE/CVF Winter Conference on Applications of Computer Vision*, 2025.

[26] Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, 2022.

[27] Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo-Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, and Ludwig Schmidt. Robust fine-tuning of zero-shot models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.

A Additional Geometric Intuition

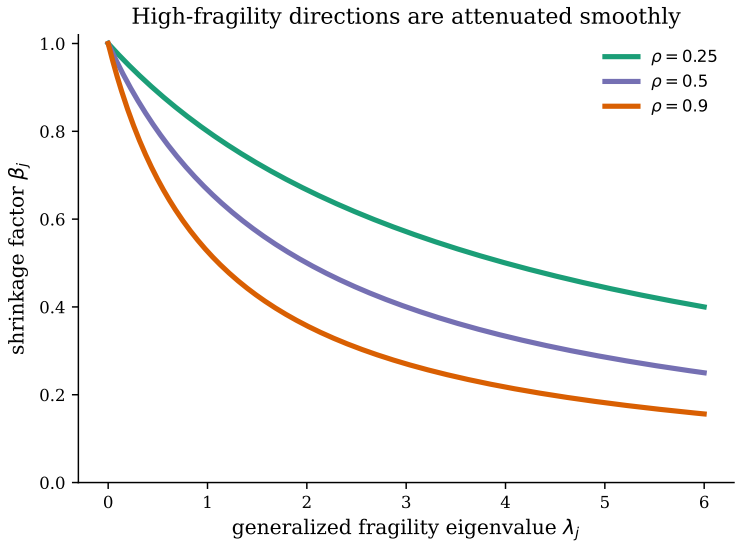


Figure 6: **Regularization path.** Each generalized direction is shrunk by $\beta_j(\lambda) = 1/(1 + \lambda\nu_j)$. Larger fragility eigenvalues decay faster as λ grows.

B Proofs

B.1 Proof of Theorem 1

Proof. Fix $z \in \mathcal{Z}$ and $Q \in \mathcal{Q}(\varrho)$. By Assumption 3, there exists h such that $dQ/dP = 1 + h$, $\mathbb{E}_P[h] = 0$, and $\mathbb{E}_P[h^2] \leq \varrho$. Therefore

$$\begin{aligned} L_Q(z) - L_Q(0) &= \mathbb{E}_Q\left[\ell(\theta(z); Z) - \ell(\theta(0); Z)\right] \\ &= \mathbb{E}_P\left[(1 + h(Z))\left(\ell(\theta(z); Z) - \ell(\theta(0); Z)\right)\right]. \end{aligned} \tag{1}$$

By Assumption 2,

$$\ell(\theta(z); Z) = \ell(\theta(0); Z) + \phi(Z)^\top z + r(Z, z).$$

Substituting this into (1) gives

$$L_Q(z) - L_Q(0) = \mathbb{E}_P\left[\phi(Z)^\top z\right] + \mathbb{E}_P\left[r(Z, z)\right] + \mathbb{E}_P\left[h(Z)\phi(Z)^\top z\right] + \mathbb{E}_P\left[h(Z)r(Z, z)\right]. \tag{2}$$

Write $\bar{\phi} = \mathbb{E}_P[\phi(Z)]$. Since $\mathbb{E}_P[h] = 0$,

$$\mathbb{E}_P\left[h(Z)\phi(Z)^\top z\right] = \mathbb{E}_P\left[h(Z)(\phi(Z) - \bar{\phi})^\top z\right].$$

Apply Cauchy–Schwarz:

$$\begin{aligned} \left|\mathbb{E}_P\left[h(Z)(\phi(Z) - \bar{\phi})^\top z\right]\right| &\leq \sqrt{\mathbb{E}_P[h(Z)^2]}\sqrt{\mathbb{E}_P\left[\left((\phi(Z) - \bar{\phi})^\top z\right)^2\right]} \\ &\leq \sqrt{\varrho}\sqrt{z^\top \mathbb{E}_P\left[(\phi(Z) - \bar{\phi})(\phi(Z) - \bar{\phi})^\top\right] z} \\ &= \sqrt{\varrho\, z^\top B z}. \end{aligned} \tag{3}$$

Next, use the remainder bound from Assumption 2:

$$|r(Z, z)| \leq \frac{\kappa(Z)}{2}\|z\|^2.$$

Applying Cauchy–Schwarz again,

$$\begin{aligned} |\mathbb{E}_P\left[h(Z)r(Z, z)\right]| &\leq \sqrt{\mathbb{E}_P[h(Z)^2]}\sqrt{\mathbb{E}_P[r(Z, z)^2]} \\ &\leq \sqrt{\varrho}\sqrt{\mathbb{E}_P\left[\frac{\kappa(Z)^2}{4}\|z\|^4\right]} \\ &= \frac{\sqrt{\varrho}}{2}\|z\|^2\sqrt{\mathbb{E}_P[\kappa(Z)^2]} \\ &\leq \frac{\sqrt{\varrho}\,K}{2}\|z\|^2. \end{aligned} \tag{4}$$

Finally, by Assumption 1,

$$L_P(z) - L_P(0) = a^\top z + \frac{1}{2} z^\top A_0 z.$$

The first two terms on the right-hand side of (2) therefore combine to

$$\mathbb{E}_P\left[\phi(Z)^\top z\right] + \mathbb{E}_P\left[r(Z, z)\right] = L_P(z) - L_P(0) = a^\top z + \frac{1}{2} z^\top A_0 z.$$

Combining this identity with (2), (3), and (4) yields

$$L_Q(z) - L_Q(0) \leq a^\top z + \frac{1}{2} z^\top A_0 z + \sqrt{\varrho\, z^\top B z} + \frac{\sqrt{\varrho}\,K}{2}\|z\|^2.$$

Adding $L_Q(0)$ to both sides gives the claimed inequality. □

B.2 Proof of Corollary 1

Proof. Start from Theorem 1:

$$L_Q(z) \le L_Q(0) + a^\top z + \frac{1}{2} z^\top A_0 z + \sqrt{\varrho\, z^\top B z} + \frac{\sqrt{\varrho}\, K}{2} \|z\|^2 .$$

For any $\lambda > 0$ and any $u \ge 0$, Young's inequality gives

$$\sqrt{\varrho\, u} \le \frac{\varrho}{2\lambda} + \frac{\lambda}{2} u.$$

Apply this with $u = z^\top B z$ to obtain

$$\sqrt{\varrho\, z^\top B z} \le \frac{\varrho}{2\lambda} + \frac{\lambda}{2} z^\top B z.$$

Also,

$$\frac{1}{2} z^\top A_0 z + \frac{\sqrt{\varrho}\, K}{2} \|z\|^2 \le \frac{1}{2} z^\top \left(A_0 + \tau I_r\right) z = \frac{1}{2} z^\top A_\tau z,$$

where the inequality uses $\tau \ge \sqrt{\varrho} K$. Substituting both identities into the bound from Theorem 1 yields

$$L_Q(z) \le L_Q(0) + \frac{\varrho}{2\lambda} + a^\top z + \frac{1}{2} z^\top A_\tau z + \frac{\lambda}{2} z^\top B z.$$

This is the desired claim. □

B.3 Proof of Theorem 2

Proof. Because $A_\tau \succ 0$ and $B \succeq 0$, the matrix $A_\tau + \lambda B$ is positive definite for every $\lambda > 0$. Hence the quadratic function

$$J_{\lambda,\tau}(z) = a^\top z + \frac{1}{2} z^\top A_\tau z + \frac{\lambda}{2} z^\top B z$$

is strongly convex and has a unique minimizer. Its gradient is

$$\nabla J_{\lambda,\tau}(z) = a + A_\tau z + \lambda B z.$$

Setting the gradient equal to zero yields the normal equation

$$(A_\tau + \lambda B) z = -a.$$

Therefore

$$z^{\mathrm{unc}}_{\lambda,\tau} = -(A_\tau + \lambda B)^{-1} a.$$

Because $z_\tau = -A_\tau^{-1} a$, we may rewrite this as

$$z^{\mathrm{unc}}_{\lambda,\tau} = (A_\tau + \lambda B)^{-1} A_\tau z_\tau.$$

This proves the first claim.

For the second claim, because $A_\tau \succ 0$ and $B \succeq 0$, the symmetric generalized eigenproblem admits an A_τ-orthonormal basis $V = [v_1, \dots, v_r]$ and nonnegative eigenvalues $\nu_1, \dots, \nu_r$ satisfying

$$BV = A_\tau V \operatorname{diag}(\nu_1, \dots, \nu_r), \qquad V^\top A_\tau V = I.$$

Expand

$$z_\tau = Vc, \qquad c = V^\top A_\tau z_\tau.$$

Since $z = Vy$ if and only if $y = V^\top A_\tau z$, we may write

$$\begin{aligned} J_{\lambda,\tau}(Vy) &= a^\top V y + \frac{1}{2} y^\top V^\top A_\tau V y + \frac{\lambda}{2} y^\top V^\top B V y \\ &= a^\top V y + \frac{1}{2} y^\top y + \frac{\lambda}{2} y^\top \operatorname{diag}(\nu_1, \dots, \nu_r) y. \end{aligned} \tag{5}$$

Now use $a = -A_\tau z_\tau = -A_\tau V c$, so

$$a^\top V y = -c^\top V^\top A_\tau V y = -c^\top y.$$

Substituting this into (5) gives

$$J_{\lambda,\tau}(Vy) = \sum_{j=1}^{r} \left[-c_j y_j + \frac{1}{2} y_j^2 + \frac{\lambda}{2} \nu_j y_j^2 \right].$$

The coordinates decouple. Minimizing each coordinate separately yields

$$0 = -c_j + (1 + \lambda \nu_j) y_j, \qquad y_j = \frac{c_j}{1 + \lambda \nu_j}.$$

Therefore

$$z_{\lambda,\tau}^{\mathrm{unc}} = Vy = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j.$$

This is exactly the stated generalized eigen-shrinkage formula. □

B.4 Proof of Proposition 1

Proof. The family $\mathcal{F}_{\mathrm{sc}} = \{\alpha z_\tau : \alpha \in \mathbb{R}\}$ is a subset of $\mathbb{R}^r$. Since $z_{\lambda,\tau}^{\mathrm{unc}}$ is the global minimizer of $J_{\lambda,\tau}$ over $\mathbb{R}^r$,

$$\min_{\alpha \in \mathbb{R}} J_{\lambda,\tau}(\alpha z_\tau) \geq J_{\lambda,\tau}(z_{\lambda,\tau}^{\mathrm{unc}}).$$

This proves the first inequality.

It remains to characterize equality. By Theorem 2,

$$z_{\lambda,\tau}^{\mathrm{unc}} = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j.$$

Equality holds if and only if $z_{\lambda,\tau}^{\mathrm{unc}} \in \mathcal{F}_{\mathrm{sc}}$, that is, if and only if there exists $\alpha \in \mathbb{R}$ such that

$$\sum_{j=1}^{r} \frac{c_j}{1 + \lambda \nu_j} v_j = \alpha \sum_{j=1}^{r} c_j v_j.$$

Because the generalized eigenvectors form a basis, this is equivalent to

$$\frac{c_j}{1 + \lambda \nu_j} = \alpha c_j \qquad \text{for all } j = 1, \ldots, r.$$

For every index j such that $c_j \neq 0$, the condition becomes

$$\alpha = \frac{1}{1 + \lambda \nu_j}.$$

Hence the same scalar α can satisfy all active coordinates if and only if

$$\nu_j = \nu_k \qquad \text{for all } j, k \text{ with } c_j \neq 0,\ c_k \neq 0.$$

Conversely, if all active generalized eigenvalues are equal to some common value $\bar{\nu}$, then

$$z_{\lambda,\tau}^{\mathrm{unc}} = \sum_{j=1}^{r} \frac{c_j}{1 + \lambda \bar{\nu}} v_j = \frac{1}{1 + \lambda \bar{\nu}} z_\tau,$$

which belongs to $\mathcal{F}_{\mathrm{sc}}$. Therefore equality holds if and only if the generalized eigenvalues are constant on the support of z_τ. □

B.5 Proof of Proposition 2

Proof. Because $z_{\lambda,\tau}^{\mathrm{unc}}$ is the minimizer of the quadratic $J_{\lambda,\tau}$, the standard completion-of-squares identity around the minimizer gives

$$J_{\lambda,\tau}(z) = J_{\lambda,\tau}(z_{\lambda,\tau}^{\mathrm{unc}}) + \frac{1}{2}(z - z_{\lambda,\tau}^{\mathrm{unc}})^\top (A_\tau + \lambda B)(z - z_{\lambda,\tau}^{\mathrm{unc}}) \qquad \text{for all } z.$$

Substituting $z = z_\tau$ yields

$$J_{\lambda,\tau}(z_\tau) = J_{\lambda,\tau}(z^{\mathrm{unc}}_{\lambda,\tau}) + \frac{1}{2}(z_\tau - z^{\mathrm{unc}}_{\lambda,\tau})^\top (A_\tau + \lambda B)(z_\tau - z^{\mathrm{unc}}_{\lambda,\tau}).$$

Rearranging gives the claimed identity.

The difference is zero if and only if $z_\tau = z^{\mathrm{unc}}_{\lambda,\tau}$. From Theorem 2,

$$z^{\mathrm{unc}}_{\lambda,\tau} = (A_\tau + \lambda B)^{-1} A_\tau z_\tau.$$

Hence $z^{\mathrm{unc}}_{\lambda,\tau} = z_\tau$ if and only if

$$(A_\tau + \lambda B) z_\tau = A_\tau z_\tau,$$

which is equivalent to $\lambda B z_\tau = 0$, hence $B z_\tau = 0$. This proves the second statement. □

B.6 Proof of Theorem 3

Proof. Let

$$M = A_\tau + \lambda B, \qquad \hat{A}_\tau = \hat{A}_0 + \tau I_r, \qquad \hat{M} = \hat{A}_\tau + \lambda \hat{B}.$$

By definition,

$$\left\| \hat{M} - M \right\| \le \left\| \hat{A}_\tau - A_\tau \right\| + \lambda \left\| \hat{B} - B \right\| = \delta.$$

Since $M \succ 0$ with $\lambda_{\min}(M) = m$ and $\delta < m$, Weyl's inequality implies

$$\lambda_{\min}(\hat{M}) \ge \lambda_{\min}(M) - \left\| \hat{M} - M \right\| \ge m - \delta > 0.$$

Therefore $\hat{M}$ is positive definite, and

$$\left\| \hat{M}^{-1} \right\| \le \frac{1}{m - \delta}.$$

Now use the exact normal equations

$$M z^{\mathrm{unc}}_{\lambda,\tau} = -a, \qquad \hat{M} \hat{z}_{\lambda,\tau} = -\hat{a}.$$

Subtract the first from the second:

$$\hat{M}(\hat{z}_{\lambda,\tau} - z^{\mathrm{unc}}_{\lambda,\tau}) = -(\hat{a} - a) - (\hat{M} - M) z^{\mathrm{unc}}_{\lambda,\tau}.$$

Multiply by $\hat{M}^{-1}$:

$$\hat{z}_{\lambda,\tau} - z^{\mathrm{unc}}_{\lambda,\tau} = -\hat{M}^{-1}(\hat{a} - a) - \hat{M}^{-1}(\hat{M} - M) z^{\mathrm{unc}}_{\lambda,\tau}.$$

Taking norms and using the bound on $\hat{M}^{-1}$,

$$\begin{aligned} \left\| \hat{z}_{\lambda,\tau} - z^{\mathrm{unc}}_{\lambda,\tau} \right\| &\le \left\| \hat{M}^{-1} \right\| \left[\|\hat{a} - a\| + \left\| \hat{M} - M \right\| \, \left\| z^{\mathrm{unc}}_{\lambda,\tau} \right\| \right] \\ &\le \frac{\|\hat{a} - a\| + \delta \left\| z^{\mathrm{unc}}_{\lambda,\tau} \right\|}{m - \delta}. \end{aligned}$$

This proves the deterministic perturbation inequality.

If $\|\hat{a} - a\| \to 0$, $\hat{A}_0 \to A_0$, and $\hat{B} \to B$, then $\delta \to 0$ and the right-hand side converges to zero, implying $\hat{z}_{\lambda,\tau} \to z^{\mathrm{unc}}_{\lambda,\tau}$. □

B.7 Proof of Corollary 2

Proof. Corollary 1 states that for every $z \in \mathcal{Z}$,

$$L_Q(z) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_{\lambda,\tau}(z).$$

Apply this inequality at $z = z^{\mathcal{Z}}_{\lambda,\tau}$:

$$L_Q(z^{\mathcal{Z}}_{\lambda,\tau}) \le L_Q(0) + \frac{\varrho}{2\lambda} + J_{\lambda,\tau}(z^{\mathcal{Z}}_{\lambda,\tau}).$$

Because $z_{\lambda,\tau}^{\mathcal{Z}}$ minimizes $J_{\lambda,\tau}$ over $\mathcal{Z}$,

$$J_{\lambda,\tau}(z_{\lambda,\tau}^{\mathcal{Z}}) \leq J_{\lambda,\tau}(z) \qquad \text{for all } z \in \mathcal{Z}.$$

Combining the two inequalities yields

$$L_Q(z_{\lambda,\tau}^{\mathcal{Z}}) \leq L_Q(0) + \frac{\varrho}{2\lambda} + J_{\lambda,\tau}(z_{\lambda,\tau}^{\mathcal{Z}}) \leq L_Q(0) + \frac{\varrho}{2\lambda} + J_{\lambda,\tau}(z) \qquad \text{for all } z \in \mathcal{Z}.$$

This proves the corollary. □

C Optional Numerical Properties

Although SWING has a closed-form solution, one useful limit statement is worth recording.

Proposition 3 (Shrinkage path limits)**.** *Under the assumptions of Theorem 2, let*

$$z_{\lambda,\tau}^{\text{unc}} = \sum_{j=1}^{r} \frac{c_j}{1+\lambda\nu_j} v_j.$$

Then:

1. $z_{\lambda,\tau}^{\text{unc}} \to z_\tau$ *as* $\lambda \downarrow 0$*;*
2. $z_{\lambda,\tau}^{\text{unc}} \to \sum_{\nu_j=0} c_j v_j$ *as* $\lambda \to \infty$*;*
3. *for every* j *with* $\nu_j > 0$*, the coefficient* $\frac{c_j}{1+\lambda\nu_j}$ *is continuous in* λ *and strictly decreases in magnitude as* λ *increases.*

Proof. All three claims are immediate from the closed-form coefficient

$$\frac{c_j}{1+\lambda\nu_j}.$$

If $\lambda \downarrow 0$, then $1 + \lambda\nu_j \to 1$, so each coefficient converges to c_j. If $\lambda \to \infty$, then the coefficient converges to 0 for every $\nu_j > 0$ and remains c_j for every $\nu_j = 0$. The continuity and monotonicity follow from elementary calculus because $\nu_j \geq 0$. □